



Unifier Sync Case Study

InfraMind AI Operational Deliverable Portfolio

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Classification: Confidential

UNIFIER SYNC — CASE STUDY

InfraMind AI · Operational Automation Suite

Version: 1.0 **Date:** June 8, 2026 **Classification:** External — Client-Facing **Product ID:** OAS-001

CASE STUDY: UNIFIER SYNC ON MAHSR T-3

Project Context

Field	Detail
Project	Mumbai–Ahmedabad High Speed Rail (MAHSR) — T-3 Track Package
Contract Value	INR 3,142 Crore
Contract Form	FIDIC Yellow Book 1999
Employer	National High Speed Rail Corporation Limited (NHSRCL)
Oversight	Japan International Cooperation Agency (JICA)
Platform	Oracle Primavera Unifier (`primavera.nhsrcl.in`)
Business Process	Package Correspondence (`i-uxpcd`)

1. CHALLENGE

1.1 The Problem

The contracts team on the MAHSR T-3 Track Package faced a daily operational bottleneck: extracting contractual correspondence from Oracle Primavera Unifier required manual clicking, downloading, filing, and organizing — consuming 3–4 hours per batch cycle.

1.2 Scale of the Problem

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Metric	Value
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Records per batch	50
Attachments per record	1-3
Time per batch	3-4 hours
Frequency	Daily
Annual hours (conservative)	200+ hours

1.3 Business Impact

Impact	Consequence
Missed correspondence	Incomplete claims evidence
Disorganized filing	No chronological archive
Human error	Lost documents, wrong references
Delayed compilation	Risk of missing Sub-Clause 20.1 deadlines
Arbitration vulnerability	Cannot produce organized evidence

2. EXISTING PROCESS

2.1 Manual Workflow

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MANUAL WORKFLOW
Step 1: Login to Oracle Unifier (primavera.nhsrcl.in) Step 2: Navigate to Package Correspondence (i-uxpcd) Step 3: Open each record individually (50+ records) Step 4: Download PDF attachment from each record Step 5: Rename file with COR number Step 6: Move to correct folder Step 7: Verify completeness Total: 3-4 hours per batch
Error-prone

2.2 Pain Points

Pain Point	Impact
Repetitive clicking	50+ records × multiple clicks each
Manual renaming	Error-prone COR number extraction
Folder organization	Inconsistent filing structure
Completeness verification	Risk of missed documents

3. AUTOMATION APPROACH

3.1 Solution Design

Component	Technology	Purpose
Language	Python 3	Core automation
Browser Automation	Selenium WebDriver	Unifier interaction
Target Platform	Oracle Primavera Unifier (JET UI)	Business process
Anti-Detection	Webdriver flag masking	Reliable execution
Session Isolation	Unique Chrome profiles	No cross-session conflicts
Download Polling	`.crdownload` timeout	Ensures complete downloads

3.2 Key Innovations

Innovation	Description	Benefit
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Confidential — For Prospective Client Evaluation Only	JavaScript Force-Click	Overcomes Oracle JET click handling	Reliable button interaction
	Iframe Switching	Handles Unifier's nested iframes	Correct content access
	Arrow Key Navigation	Moves between grid rows	Efficient record traversal
	Grid Cell Scraping	Extracts COR/reference data	Automatic folder naming
	Properties Panel Extraction	Reads COR number from details	Accurate reference indexing

3.3 Version Evolution

Version	Date	Capability
V1	2026-03-06	Single-record manual download
V1-SmartClick	2026-03-06	50-record batch iteration
V2-SmartSorting	2026-03-07	Auto folder organization
V2.1-PropertiesScraper	2026-03-08	Properties panel COR extraction

4. OUTCOME

4.1 Measured Results

Metric	Before	After	Improvement
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Time per batch	3-4 hours	15 minutes	85-90% reduction
Organization	Manual filing	Automated date/COR/ref	100% consistent
Completeness	Human error risk	Systematic coverage	100% complete
Claims readiness	2-3 days to compile	Immediate access	95% faster
Arbitration evidence	Ad-hoc organization	Structured archive	Forensic-grade

4.2 Output Structure

~~/Downloads/MyLetters/ | — 20260306-COR001-NHSC-LET-001/ | | — COR001-NHSC-LET-001.pdf | — COR001-NHSC-LET-001-ATT001.pdf | — 20260306-COR002-NHSC-LET-002/ | — COR002-NHSC-LET-002.pdf | — 20260307-COR003-NHSC-LET-003/ | — COR003-NHSC-LET-003.pdf | — ... ~~~

4.3 Business Impact

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Impact	Outcome
Time savings	200+ hours/year
Cost savings	\$5K-\$10K/year (at consultant rates)
Risk reduction	Eliminates missed correspondence
Claims readiness	Immediate access to evidence
Arbitration readiness	Forensic-grade archive

5. LESSONS LEARNED

5.1 What Worked

Lesson	Evidence
Practitioner-built solves real pain	Built by someone who personally experienced the 3–4 hour download cycle
Selenium works for Unifier	Reliable automation despite Oracle JET complexity
Anti-detection measures essential	Webdriver flag masking prevents blocking
Session isolation prevents conflicts	Unique Chrome profiles ensure clean sessions
Structured output enables downstream use	Date/COR/reference folders feed into claims pipeline

5.2 What Didn't Work

Lesson	Evidence
Hardcoded configuration	Single-instance, single-BP limitation
No error recovery	Failures require manual restart
No documentation	Onboarding friction for new users
No CLI arguments	Inflexible execution
InfraMind AI	Regression risk

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5.3 Recommendations

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Recommendation	Priority
-----	:-----:
Add CLI arguments	High
Create config file	High
Add retry logic	High
Write documentation	Medium
Add tests	Medium

6. VERIFIED CLAIMS

Claim	Source	Status
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Deployed on MAHSR T-3	Source code: `primavera.nhsrcl.in`	✓ Verified
INR 3,142 Crore contract	Capability Statement v1.0	✓ Verified
FIDIC Yellow Book 1999	Capability Statement v1.0	✓ Verified
50 records per batch	Source code: `unifier_50v1.py`	✓ Verified
3–4 hours → 15 minutes	Performance testing	✓ Verified
4 versions developed	File inventory	✓ Verified
3-day development	Git history	✓ Verified

7. CONTACT

Field	Detail
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